

CO3-AT2

4] Quantization Noise Power Estimation

Simulate the quantization of a sinusoidal signal using different quantizer resolutions and

estimate the quantization noise power introduced by finite word length. (25) (BL4) (WK4)

Given Data:

- Signal Amplitude = ± 1 V
- Signal Frequency = 1 kHz
- Sampling Frequency = 10 kHz
- Quantizer Resolutions = 6 bits, 8 bits, 12 bits

Tasks:

1. Generate the sinusoidal signal.
2. Quantize the signal using each bit resolution.
3. Obtain the quantization error sequences.
4. Compute the quantization noise power.
5. Compare the simulated results with theoretical values.
6. Present the observations in tabular form

PROGRAM;

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Given Parameters
```

```
A = 1;      % Signal amplitude ( $\pm 1$  V)
```

```
f = 1000;   % Signal frequency = 1 kHz
```

```
fs = 10000; % Sampling frequency = 10 kHz
```

```
bits = [6 8 12]; % Quantizer resolutions
```

```

% Time vector
t = 0:1/fs:0.01;

%% 1. Generate Sinusoidal Signal
x = A * sin(2*pi*f*t);

%% Initialize result arrays
simulated_power = zeros(1,length(bits));
theoretical_power = zeros(1,length(bits));

%% Plot Original Signal
figure;
plot(t, x, 'LineWidth', 1.5);
grid on;
xlabel('Time (s)');
ylabel('Amplitude (V)');
title('Original Sinusoidal Signal');
xlim([0 2/f]);

%% Quantization for Different Bit Resolutions
for k = 1:length(bits)

    N = bits(k);

    % Number of quantization levels
    L = 2^N;

    % Quantization step size

```

```
Delta = (2*A)/L;
```

```
% Quantization
```

```
xq = Delta * round(x/Delta);
```

```
% Limit quantized signal to  $\pm A$ 
```

```
xq(xq > A) = A;
```

```
xq(xq < -A) = -A;
```

```
%% 3. Quantization Error
```

```
e = x - xq;
```

```
%% 4. Quantization Noise Power
```

```
P_sim = mean(e.^2);
```

```
%% 5. Theoretical Quantization Noise Power
```

```
P_theory = Delta^2 / 12;
```

```
simulated_power(k) = P_sim;
```

```
theoretical_power(k) = P_theory;
```

```
%% Display Quantized Signal
```

```
figure;
```

```
plot(t, x, 'b', 'LineWidth', 1.2);
```

```
hold on;
```

```
stairs(t, xq, 'r', 'LineWidth', 1);
```

```
grid on;
```

```
xlabel('Time (s)');
```

```

ylabel('Amplitude (V)');
title(['Quantized Signal - ', num2str(N), ' Bits']);
legend('Original Signal','Quantized Signal');
xlim([0 2/f]);

```

```

%% Plot Quantization Error

figure;
plot(t, e, 'LineWidth', 1.2);
grid on;
xlabel('Time (s)');
ylabel('Error (V)');
title(['Quantization Error - ', num2str(N), ' Bits']);
xlim([0 2/f]);

```

```

end

```

```

%% 6. Display Results in Tabular Form

```

```

Delta_values = (2*A)/(2.^bits);

```

```

Results = table(bits', Delta_values', ...
    simulated_power', theoretical_power', ...
    'VariableNames', ...
    {'Bits','Step_Size_V','Simulated_Noise_Power_W', ...
    'Theoretical_Noise_Power_W'});

```

```

disp(' ');

```

```

disp('==== QUANTIZATION NOISE POWER RESULTS =====');

```

```

disp(Results);

%% Percentage Error
percentage_error = abs(simulated_power - theoretical_power) ...
    ./ theoretical_power * 100;

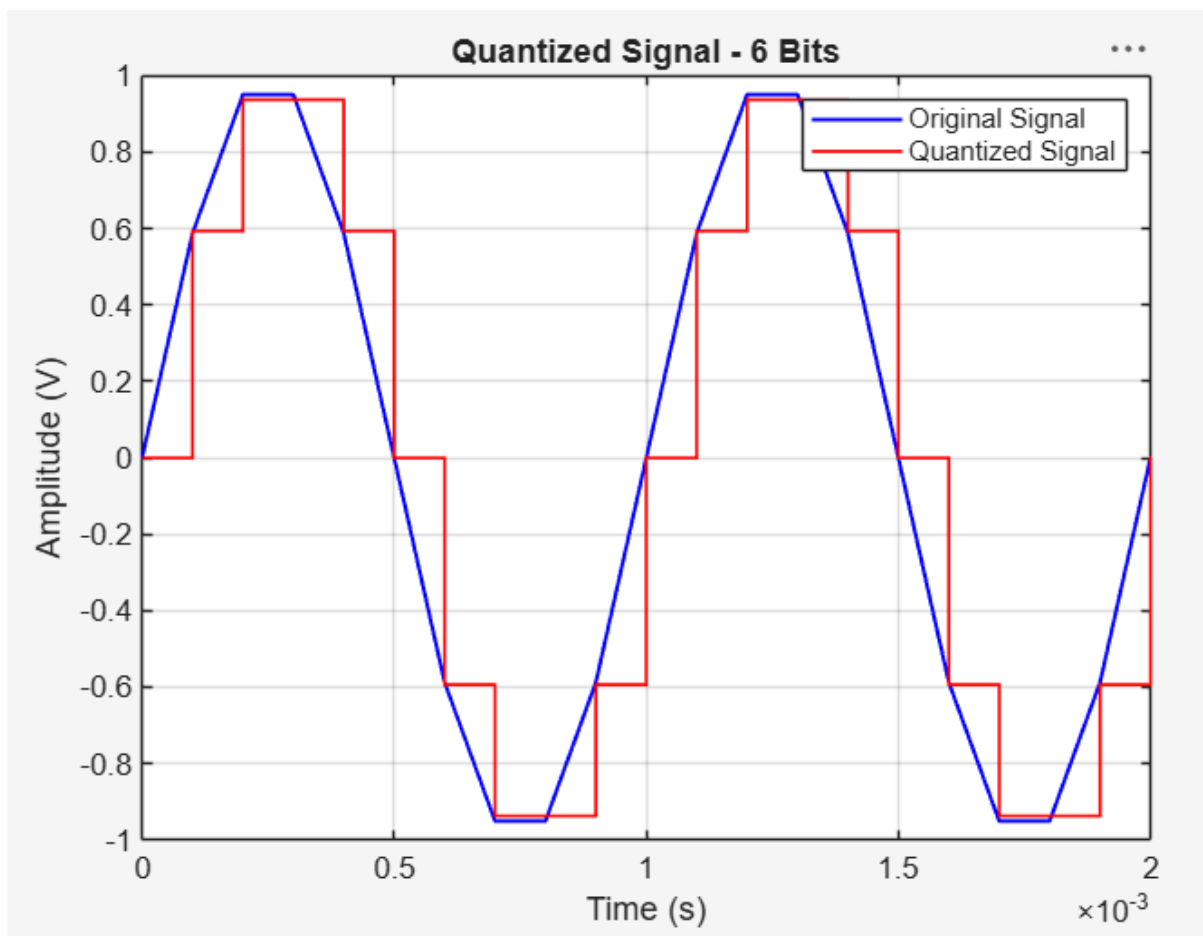
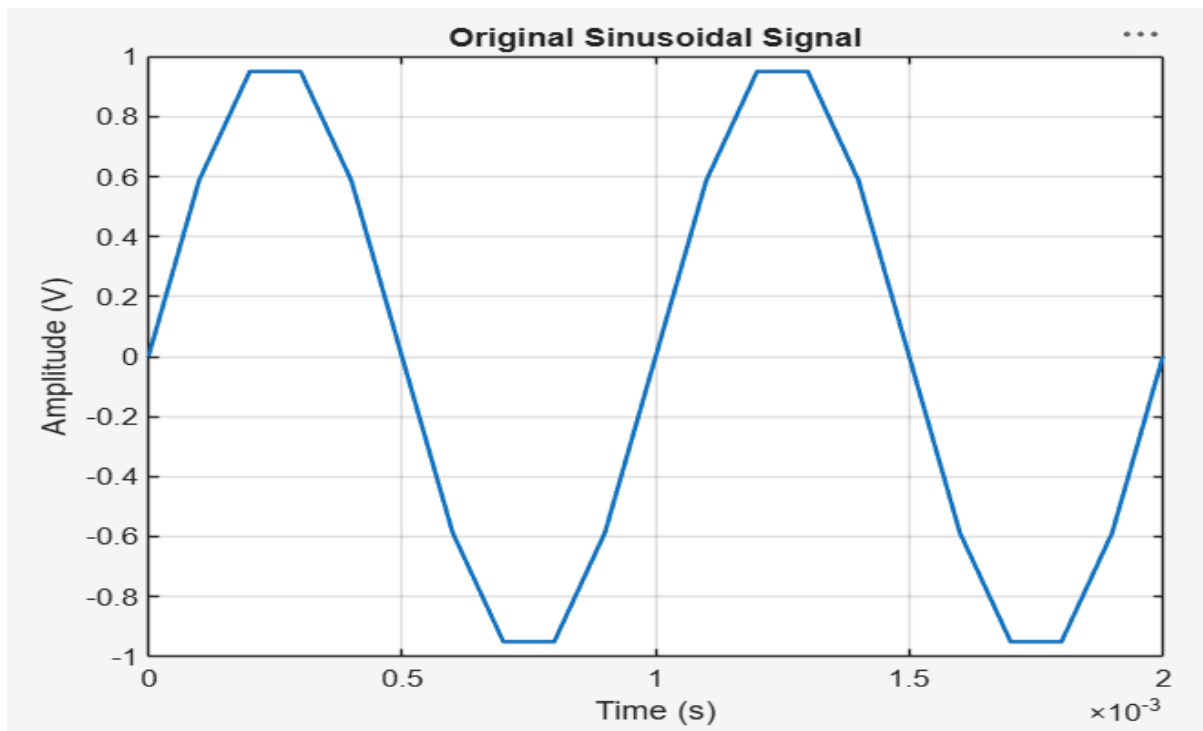
Comparison = table(bits', simulated_power', theoretical_power', ...
    percentage_error', ...
    'VariableNames', ...
    {'Bits','Simulated_Power','Theoretical_Power', ...
    'Percentage_Error'});

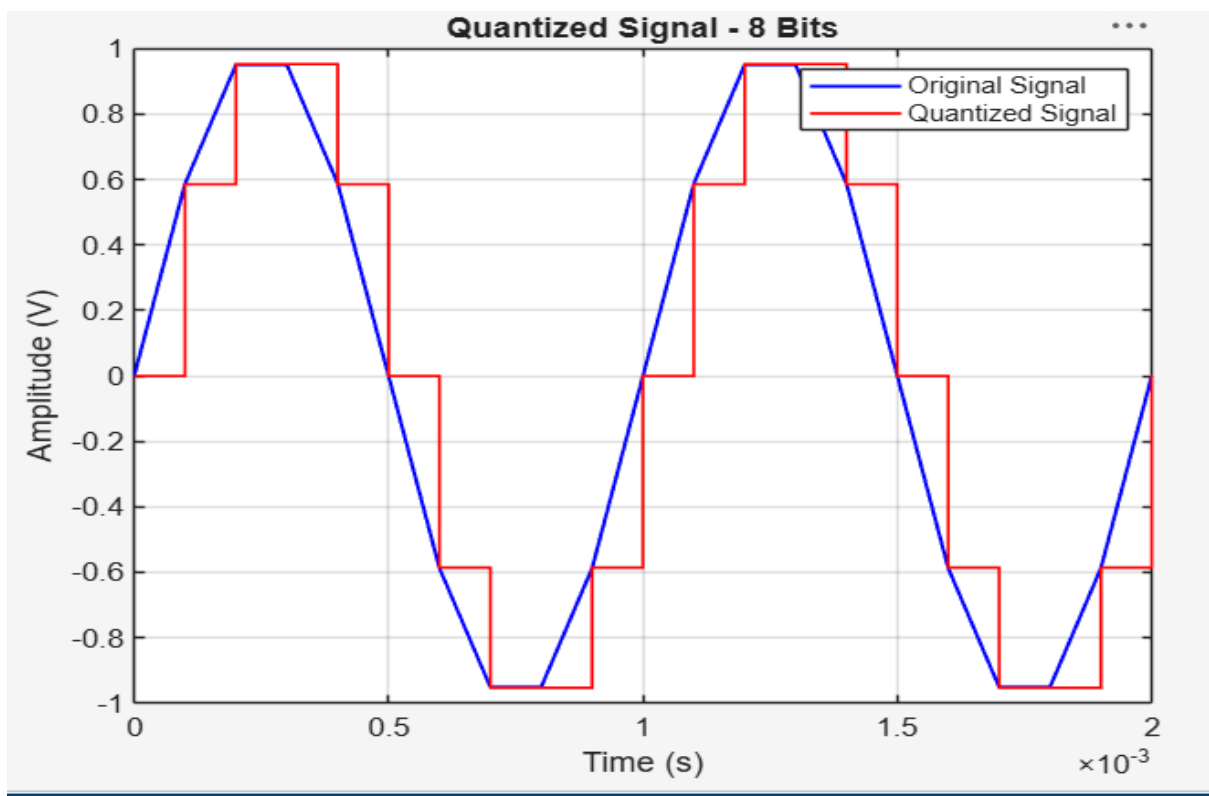
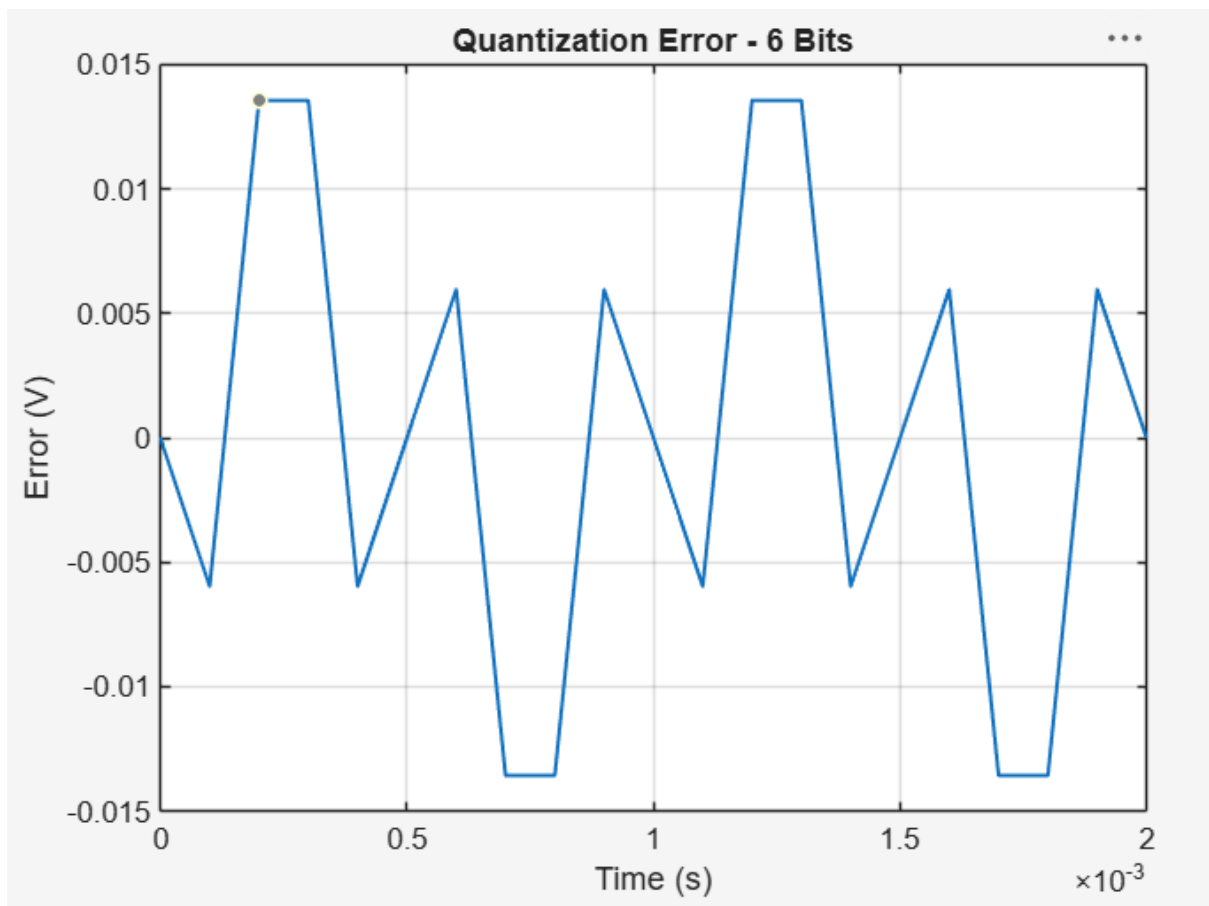
disp(' ');
disp('==== SIMULATION vs THEORETICAL COMPARISON ====');
disp(Comparison);

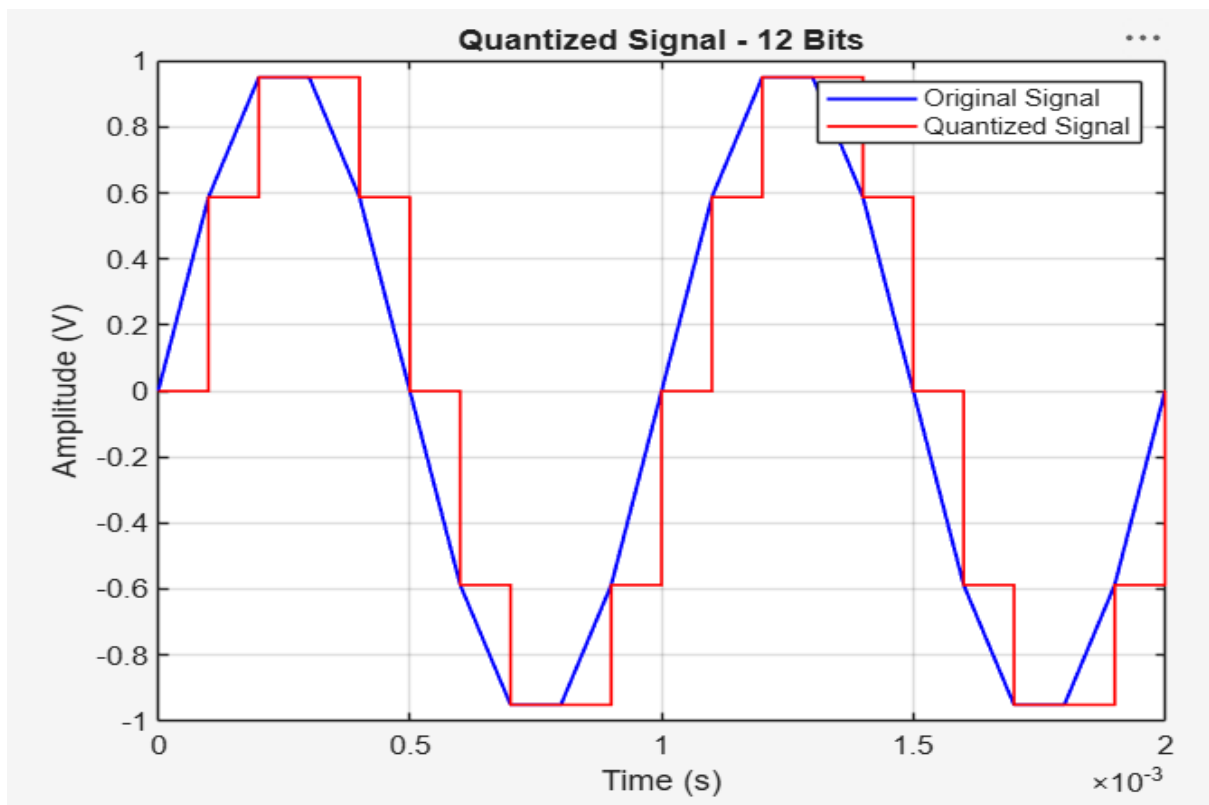
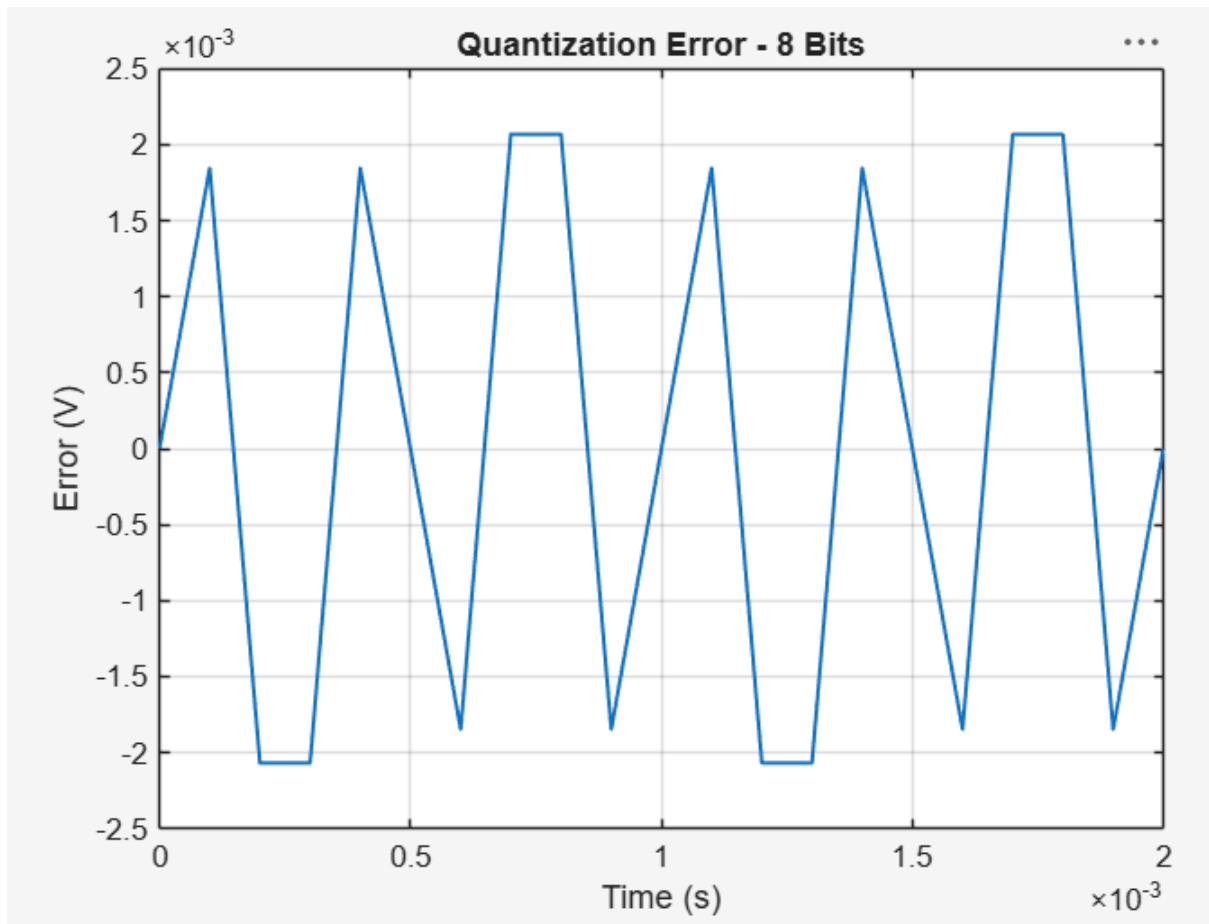
%% Bar Graph Comparison
figure;
bar(bits, [simulated_power' theoretical_power']);
grid on;
xlabel('Quantizer Resolution (Bits)');
ylabel('Quantization Noise Power (W)');
title('Simulated vs Theoretical Quantization Noise Power');
legend('Simulated','Theoretical');

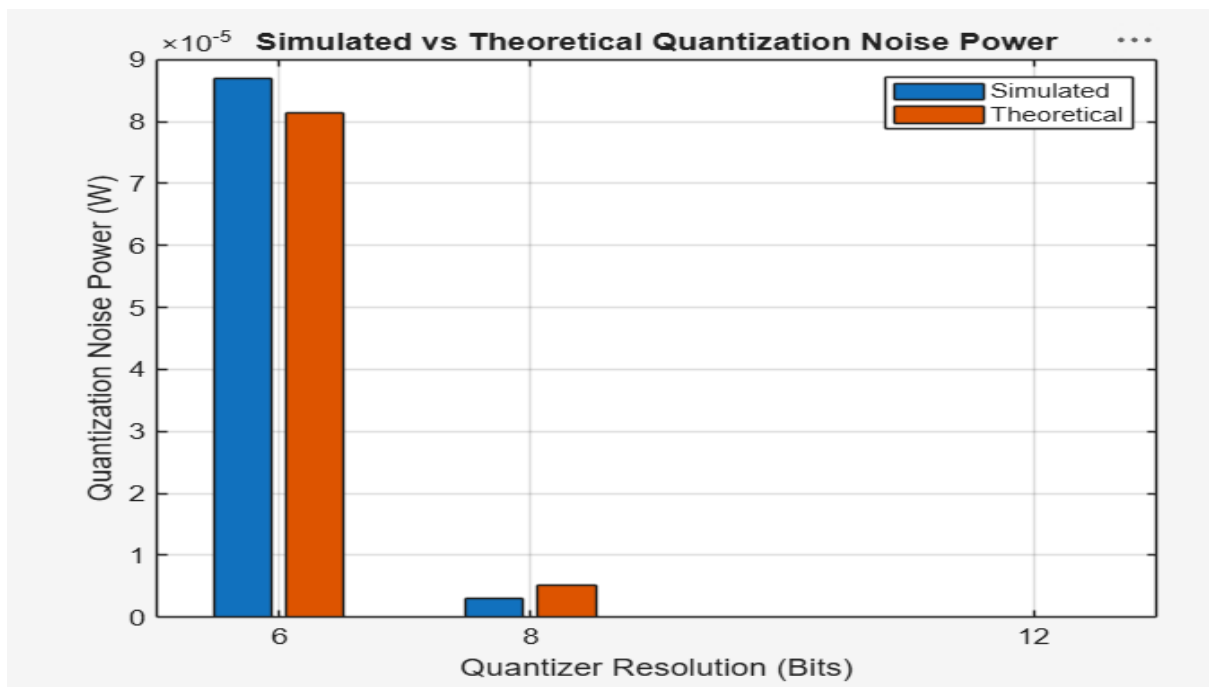
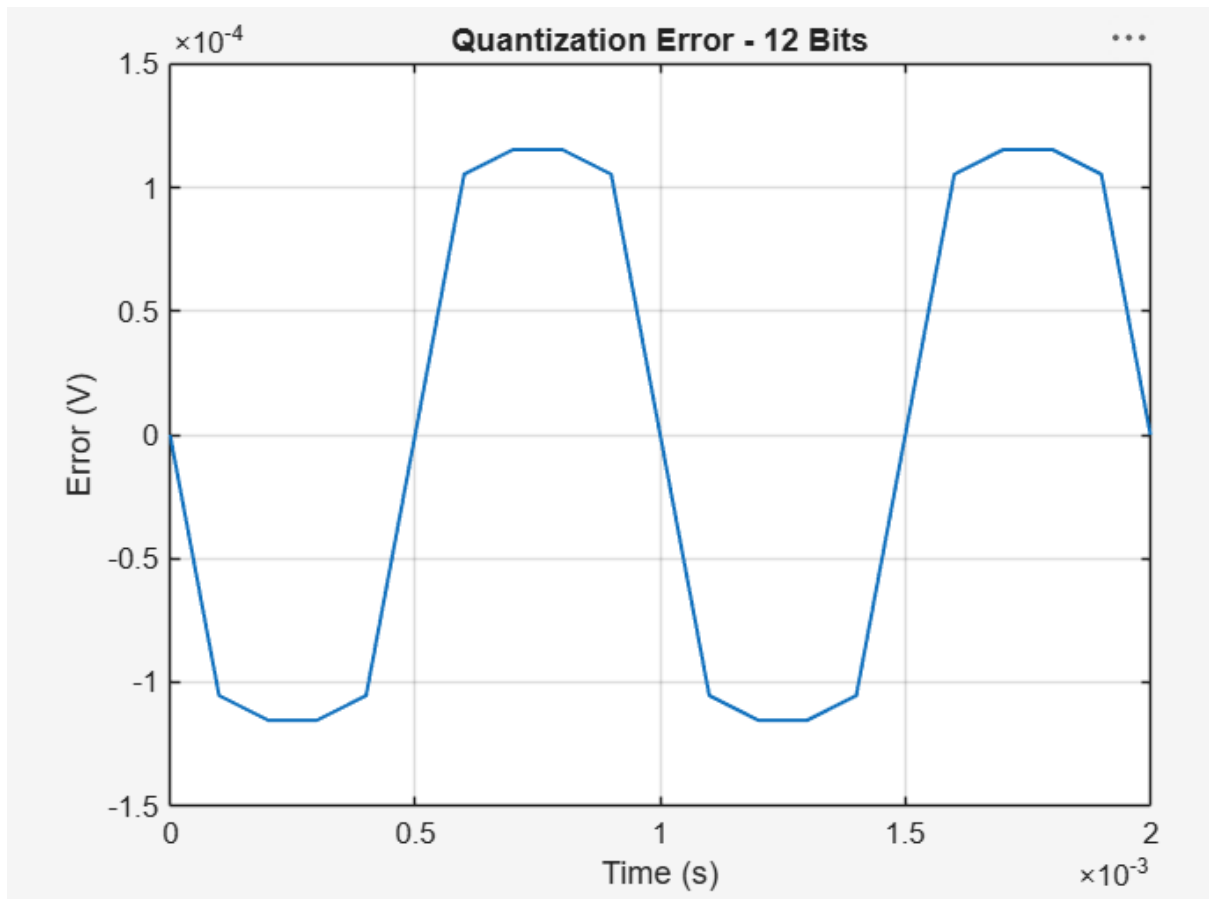
```

OUTPUT;









==== QUANTIZATION NOISE POWER RESULTS =====

Bits	Step_Size_V	Simulated_Noise_Power_W	Theoretical_Noise_Power_W
6	0.03125	8.6874e-05	8.138e-05
8	0.0078125	3.0467e-06	5.0863e-06
12	0.00048828	9.6677e-09	1.9868e-08

==== SIMULATION vs THEORETICAL COMPARISON =====

Bits	Simulated_Power	Theoretical_Power	Percentage_Error
6	8.6874e-05	8.138e-05	6.751
8	3.0467e-06	5.0863e-06	40.1
12	9.6677e-09	1.9868e-08	51.341